q1 - Write two Hadoop commands for CSUEB Hadoop to
1) create a new sub-directory called ageinput in the HDFS directory /home/user66, and
2) copy the file agesample.txt from local directory /home/student66 to the ageinput directory and change the file name to agedata.txt at the same time.

A.
1) To create a new sub-directory called ageinput in the HDFS directory /home/user66:

hadoop fs -mkdir /home/user66/ageinput
This command creates the specified directory in HDFS.

2) To copy the file agesample.txt from local directory /home/student66 to the ageinput directory and rename it to agedata.txt:

hadoop fs -put /home/student66/agesample.txt /home/user66/ageinput/agedata.txt
This command copies the local file to HDFS with the new filename in one step

q2 - Write two Hadoop commands for CSUEB Hadoop to
1) compile three Java files: AgeFinder.java, AgeFinderMapper.java, and AgeFinderReducer.java, and 2) create a Jar file for these three Java class files. Note that
a) these three Java files are in the default local directory: /home/student66,
b) AgeFinder.java is the main class, i.e., holds the main method,
c) the new jar file should be named as age-finder.jar, and
d) hadoop-common-2.6.1.jar, hadoop-mapreduce-client-core-2.6.1.jar, and commons-cli-2.0.jar are needed for compilation, and they are located in the local directory /home/student66/jarforcompile.

A.
To compile the Java files and create a JAR file for Hadoop, you'll need the following commands:

To compile the three Java files with the required dependencies:

text
javac -classpath .:/home/student66/jarforcompile/hadoop-common-2.6.1.jar:/home/student66/jarforcompile/hadoop-mapreduce-client-core-2.6.1.jar:/home/student66/jarforcompile/commons-cli-2.0.jar -d . /home/student66/AgeFinder.java /home/student66/AgeFinderMapper.java /home/student66/AgeFinderReducer.java

To create a JAR file with AgeFinder as the main class:
jar -cvf age-finder.jar -c /home/student55 .

q3 - Write two Hadoop commands to
1) run the age-finder.jar using agedata.txt in /home/user66/ageinput as the input file and using /home/user66/output1 as the directory for the output files, and
2) display the output file part-r-00000 on Hadoop screen.

A.
To run the age-finder.jar and display the output file, you'll need these two Hadoop commands:

1) To run the jar file with specified input and output directories:
hadoop jar age-finder.jar AgeFinder /home/user66/ageinput/agedata.txt /home/user66/output1
This command executes the MapReduce job in the jar file, processing the input file agedata.txt and storing results in the output1 directory.

2) To display the output file part-r-00000 on the Hadoop screen:
hdfs dfs -cat /home/user66/output1/part-r-00000

q4 - exponent = [1, 2, 3, 4, 5]

base = 3

Write a lambda function to get this result: [3, 9, 27, 81, 243]

=> result = list(map(lambda x: base**x, exponent))

q5 - data = [0, 1, 2, 3, 4, 5, 6, 7, 8, 9]
Write a lambda function to get this result: [1, 3, 5, 7, 9]

=> result = list(filter(lambda x: x % 2 != 0, data))

q6 - record = "CA94542+M22+255-78-8594"

In the above record, CA is the state (string), 94542 is a zip code (string), M stands for male (string), 22 is age (integer), 255-78-8594 is SSN (string). Write one line of Python code to assign CA to state variable, 94542 to zipcode variable, M to gender variable, 22 to age variable, and 255-78-8594 to SSN variable.

=> state, zipcode, gender, age, SSN = record[:2], record[2:7], record[8], int(record[9:11]), record[12:]

q7 - Regarding the BytesWritable on page 101, If

BytesWritable b = new BytesWritable(new byte[] { 7, 94, 14, 9 });
byte[] bytes = serialize(b);
assertThat(StringUtils.byteToHexString(bytes), is("?"));
then which of the following answer is correct for the field with the question mark?

Group of answer choices

0000000407941409

00000002075E0E09

0000000207941409

00000004075E0E09

=> 00000004075E0E09

q8 - Which of the following argument about MapReduce functions is incorrect?
Group of answer choices

Hadoop system usually automatically shuffles and sorts the Map output.

Both Map and Reduce tasks write their output to HDFS.

The output folder for a MapReduce job is always created by the Hadoop system but not by a user.

The number of Map tasks is usually determined by the size of the input data.

=> Both Map and Reduce tasks write their output to HDFS.

q9 - If a maximum temperature Map function has the output as follows:

Key (year)

Output Size for the Key

1980
300 MB

1981
200 MB

1982
100 MB

The HDFS block size is set at 80 MB. How many Reduce tasks will be created to process the Map output for the year 1980?

=> 1 reduce tasks

q10- What is the default buffer size for the output of a map function?
Group of answer choices

64 MB

100 MB

128 MB

4 KB

A. 100 MB

q11- Which of the following arguments about tasktracker failure is NOT correct?
Group of answer choices

The jobtracker always removes a failed tasktracker from tasktracker pool.

It is usually not necessary to rerun those map tasks that have already been completed successfully on a failed tasktracker.

Even if a tasktracker has not failed, the tasktracker can still be blacklisted by the jobtracker.

If a tasktracker has stopped sending heartbeats to the jobtracker, then it indicates the tasktracker fails.

A. The jobtracker always removes a failed tasktracker from tasktracker pool.

q12- String line = “7B48692351D5”;

int X = Integer.parseInt(line.substring(3, 5));

int Y = Integer.parseInt(line.substring(7, 8));

Then, X – Y =?_____

Group of answer choices

834

51

866

83

A. 83

q13- Which of the following is likely to be the default partitioner?
Group of answer choices

TotalOrderPartitioner

KeyFieldBasedPartitioner

hostPartitioner

HashPartitioner

A. HashPartitioner

q14 - Hadoop doesn’t try to diagnose and fix slow-running tasks; instead, it tries to detect when a task is running slower than expected and launches another equivalent task as a backup. This is known as _____ .
Group of answer choices

Task replica

Preventive execution of tasks

Task backup

Speculative execution of tasks

A. Speculative execution of tasks

q15 - Which of the following argument is incorrect?
Group of answer choices

Both Map and Reduce processes always have key and value as their input and output.

Reduce tasks don’t have the advantage of data locality; the input to a single reduce task is normally the output from all mappers.

For most jobs, a good split size tends to be the size of an HDFS block, 64 MB by default, although this can be changed for the cluster (for all newly created files) or specified when each file is created.

To run a Hadoop job that needs input data and generates output results, users usually upload the input data into an HDFS folder and make an output folder for the expected results.

A. Both Map and Reduce processes always have key and value as their input and output.

q16 - Which of the following object allows users to access configuration data (such as current key and value) for a Hadoop job?
Group of answer choices

FileSystem object

Path object

Context object

Job object

A. Context object

q17 - Which of the following command can NOT be used to check Spark version?
Group of answer choices

spark-sql --version

spark-shell --version

py-spark --version

spark-submit --version

A. py-spark --version

q18 - Which of the following module is needed to run a Python program on Spark?
Group of answer choices

SparkConf
SparkContext
SparkFiles
TaskContext

=> SparkContext

q19 - Which of the following Jar is needed to run a common Python Map/Reduce job on Hadoop?
Group of answer choices

hadoop-streaming-*.jar
hadoop-mapreduce-examples-*.jar
hadoop-common-*.jar
hadoop-mapreduce-client-core-*.jar

=> hadoop-streaming-*.jar

q20 - Which of the following sentences is NOT correct?
Group of answer choices

RDDs refers to Resilient Distributed Datasets, which is a collection of data objects spread across a cluster, stored in RAM or on Disk.

Transformation operation can be considered as a map() function, which is used to create new datasets from existing ones.

Action operation can be considered as a reduce() function, which runs a computation on the dataset and return results to the driver program.

A lambda function can take any number of arguments, but can only have two expressions.

=> A lambda function can take any number of arguments, but can only have two expressions.

q21- Which of the following function will create RDDs for Spark?
Group of answer choices

Collect()
Glom()
Parallelize()
Map()

=> Parallelize()

Question 112 pts
Which of the following arguments about tasktracker failure is NOT correct?
Group of answer choices

The jobtracker always removes a failed tasktracker from tasktracker pool.

It is usually not necessary to rerun those map tasks that have already been completed successfully on a failed tasktracker.

To run a Hadoop job that needs input data and generates output results, users usually upload the input data into an HDFS folder and make an output folder for the expected results.

=> The jobtracker always removes a failed tasktracker from tasktracker pool.

----------x----------x----------x----------x----------

Full lifecycle of a MapReduce job in Java:

Write `.java` files

Compile to `.class` → `javac -classpath ...`

Package into `.jar` → `jar -cvf myprogram.jar ...`

Run → `hadoop jar myprogram.jar MyDriver /input-path /output-path`

Check output in HDFS `hdfs dfs -cat /output/output-file`

Full lifecycle of a MapReduce job in Python:

Author mapper/reducer (Python) → make executable

Test locally with `echo | mapper | sort | reducer`

Put input to HDFS; clear/create output dir

Run with `hadoop jar ... hadoop-streaming.jar -mapper ... -reducer ...`

Read results with `hdfs dfs -cat .../part-r-00000`